window.*input_file = new TextDecoder('utf-8').decode(Uint8Array.from(atob('LS0tCnRpdGxlOiAiQWJzdHJhY3Rpb24gJiBDb250cmFzdCIKZm9yb
function(c) { return c.charCodeAt(0); })); window.*input_filename = '/Users/joseffruehwald/Documents/courses/Lin300-2026.github.io/course-materials/lecture-notes/15_abstraction/index.qmd'; </script>

Abstraction & Contrast

Dr. Josef Fruehwald

From Phonetics to Phonology

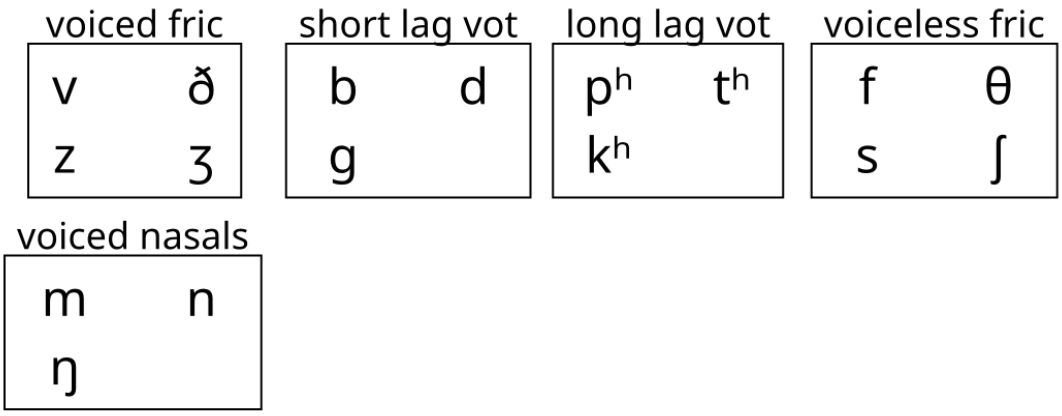
```
Loading required package: sysfonts
```

```
Loading required package: showtextdb
```

There is no definition of the difference between Phonetics and Phonology that won't make *someone* angry.

Phonetics

English Consonants



Phonology

English Consonants

+voice				-voice			
v	ð	b	d	p^h	t^h	f	θ
z	ʒ	g		k^h		s	ʃ
m	n						
ŋ							

What do you know?

Linguistics is all about answering the question “What is your knowledge of your language?”

Aside: Knowledge

Knowledge of sounds

long-lag VOT



short-lag VOT



Knowledge of Sounds

How would you transcribe your pronunciation of a novel word like:

“kowspie”

Knowledge of Sounds

	k^h	k
p^h	k^hosp^hi	$kosp^hi$

p kʰospi kospi

Knowledge of Sounds

Your preference (I'm guessing!) for [kʰospi], demonstrates a kind of knowledge about the sounds of language beyond just the fact that there are long-lag and short-lag VOT segments.

Aside: Wugs

Knowledge of Sounds

play	ˈ**p**ʰleɪ	dɪ.ˈs**p**leɪ	*display*
card	ˈ**k**ʰɑɪd	dɪ.ˈs**k**ɑɪd	*discard*

Knowledge of Sounds

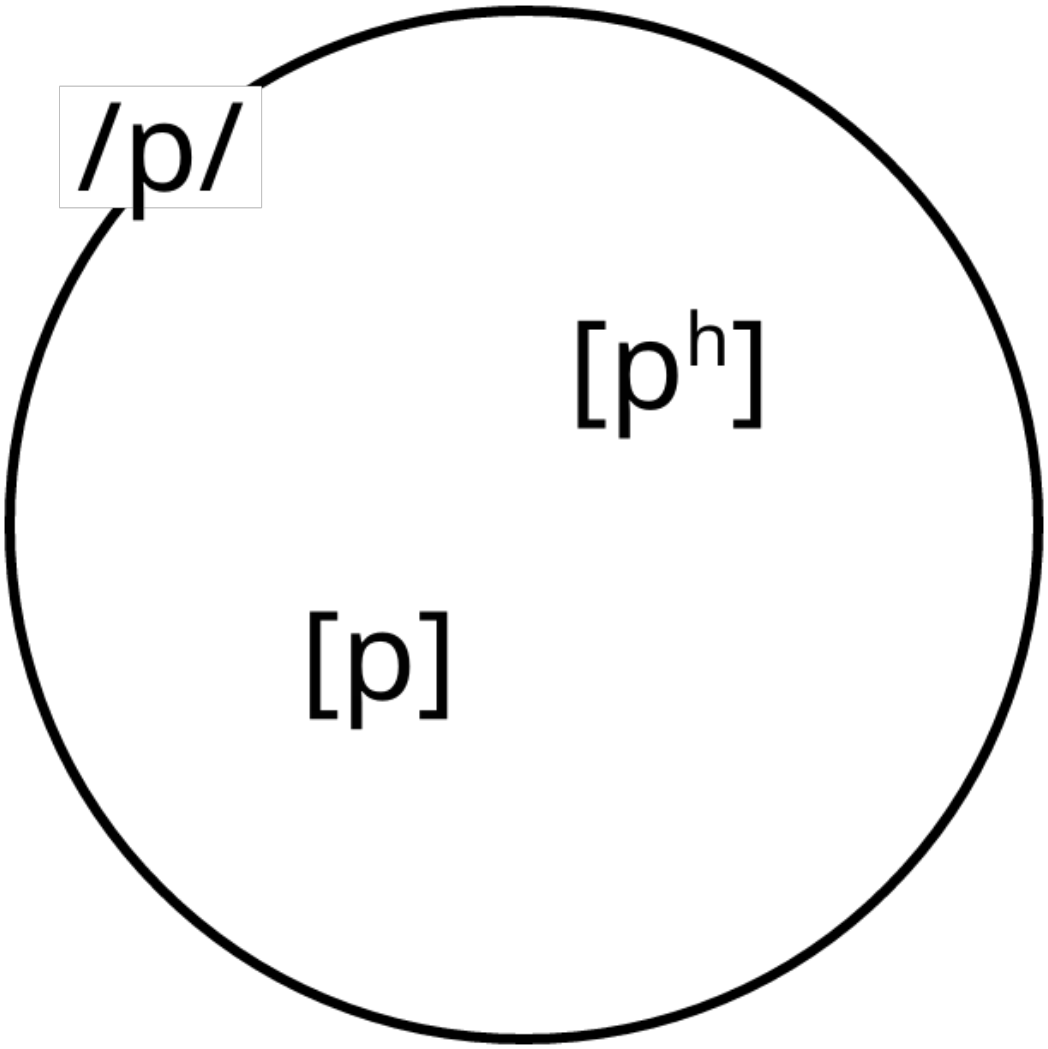
long	*longing*	*longer*
lɔŋ	lɔŋɪŋ	lɔŋgɻ

Knowledge of Sounds

Upshot: Abstraction is necessary

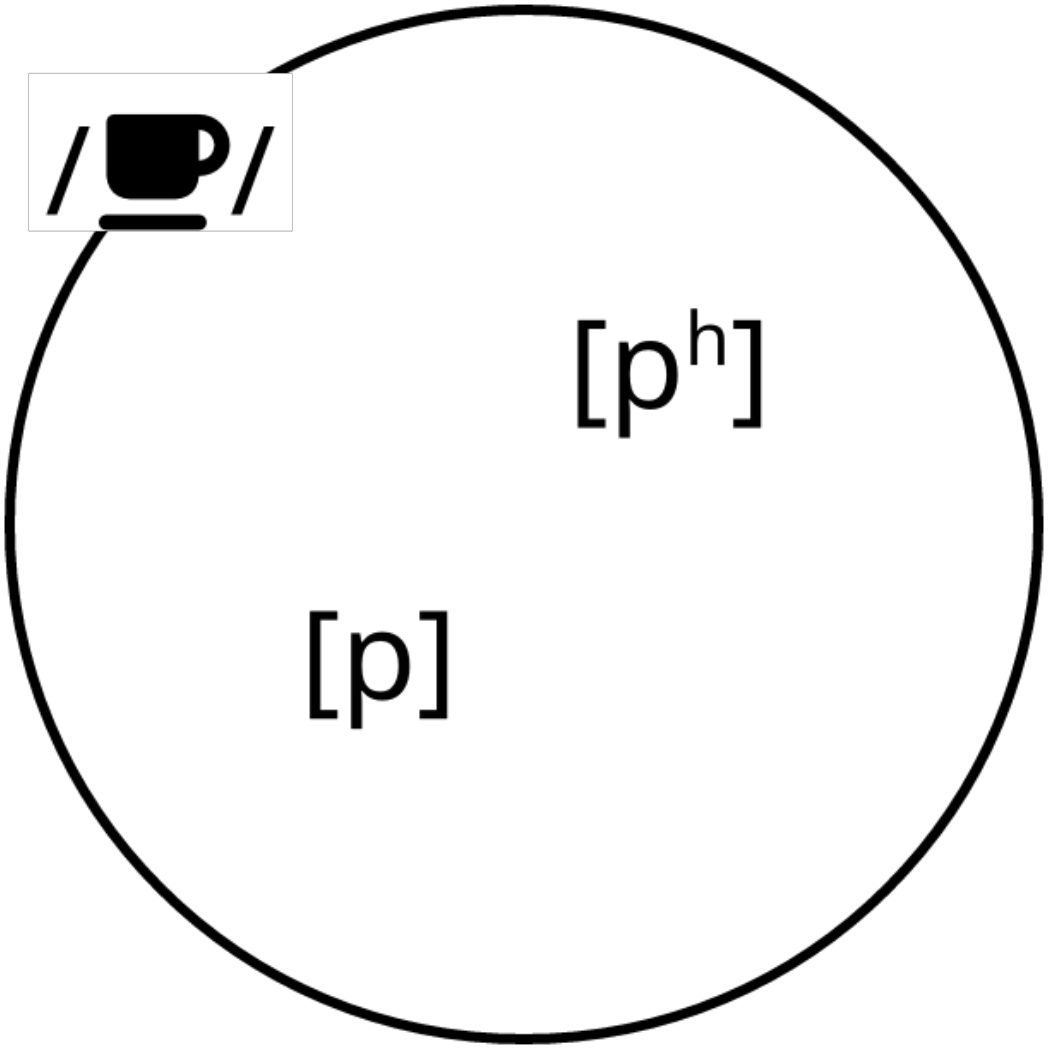
"Phoneme"

A label for a set of sounds that all count as basically the same.



The phoneme /p/ has at least two *allophones*, [p] and [pʰ]

“Phoneme”



We use “/p/” as convenient label for the phoneme, but the phoneme itself is not a sound.

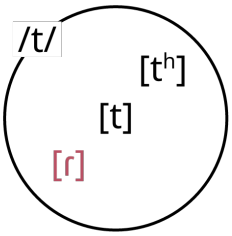
We could have used “” as the label for the phoneme, but it would be more confusing to read.

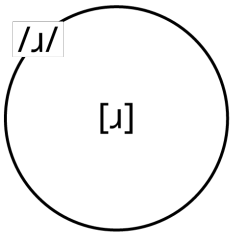
Allophones

- Not every phoneme has the same number of allophones.
- There isn’t a universal set of relationships between allophones and phonemes.

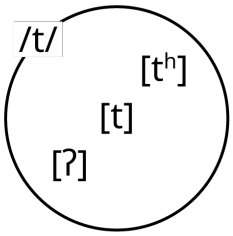
Allophones

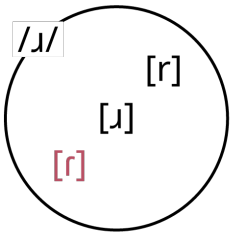
American English





Scottish English



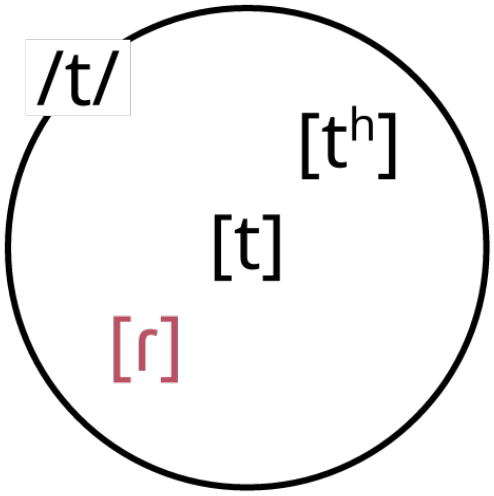


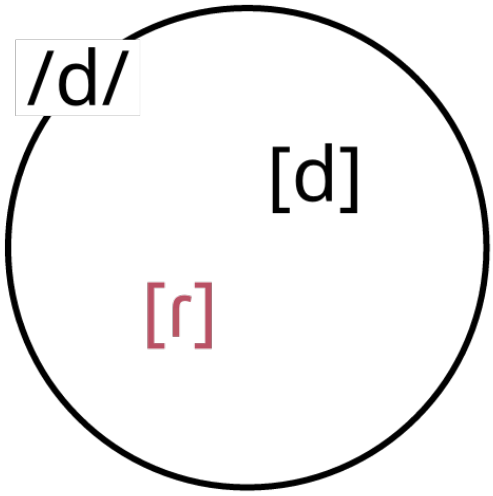
Allophones

There isn't always a 1-to-1 relationship between phones and phonemes.

Allophones

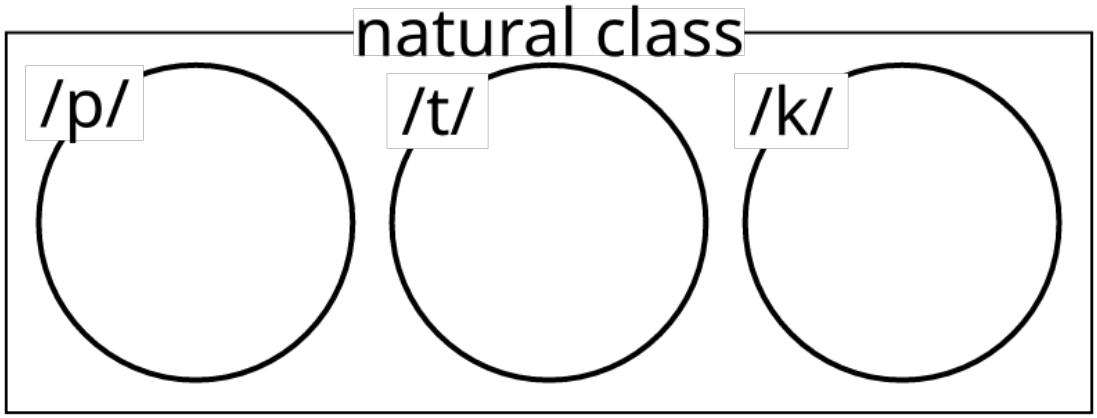
American English





"Natural Classes"

Groups of phonemes that pattern together are "natural classes"



Allophones & Natural Classes

Which sounds are allophones of which phonemes, and which phonemes form natural classes, are all language specific.

Phonemic Contrast

If two sounds are *not* basically the same, then they are *contrastive*, and are allophones of different phonemes.

ether	*either*
[iθɻ]	[iðɻ]

Phonemic Contrast

Minimal Pairs

Two words that are identical except for the one phoneme in question

d

?

ð

?

Allophonic Alternation

Complementary Distribution

If two sounds never appear in the same contexts, they are in a complementary distribution.

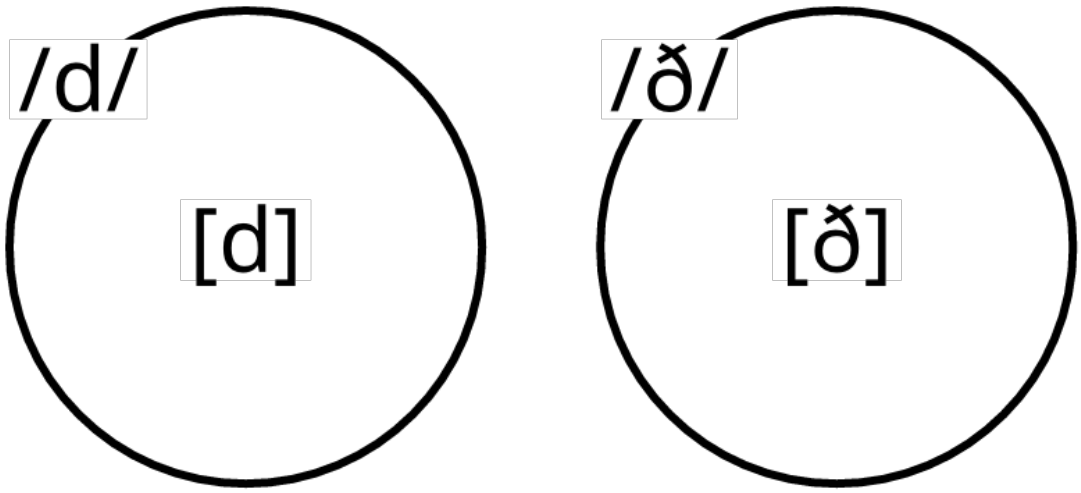
Allophonic Alternation

Spanish:

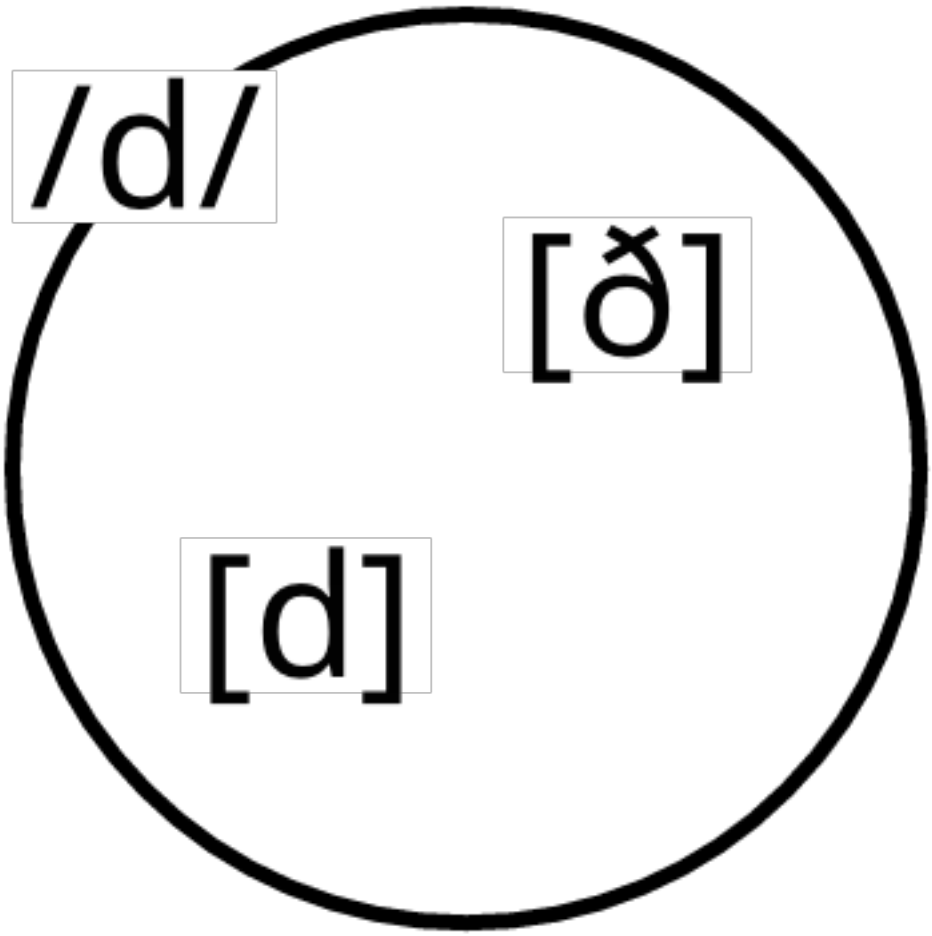
[dama]	*lady*
[dulse]	*sweet*
[disfrutas]	*disguise*
[laðo]	*side*
[universiðað]	*university*
[granaða]	*Granada*

Phonological analysis

English:



Spanish:


/d/
[ð]
[d]